SIMATS ENGINEERING

Unit 3

Hands on Exercises for Variance, Sampling,

Covariance and Correlation

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Question 1:

Mean and Population Variance

The delivery times of five online orders are: 20, 25, 30, 35, 40 minutes.

Calculate the mean and population variance.

Step 1: Identify the given data

Delivery times (X) = 20, 25, 30, 35, 40 minutes

Number of observations (N) = 5

Step 2: Calculate the sum of observations

$$\Sigma X = 20 + 25 + 30 + 35 + 40 = 150$$

Step 3: Calculate the mean

$$\text{Mean } (\mu) = \Sigma X / N$$

$$\mu = 150 / 5 = 30$$

Therefore, the mean delivery time is 30 minutes.

Step 4: Calculate deviations from the mean

X	**$X - \mu$**	**$(X - \mu)^2$**
20	$20 - 30 = -10$	100
25	$25 - 30 = -5$	25
30	$30 - 30 = 0$	0
35	$35 - 30 = 5$	25
40	$40 - 30 = 10$	100
Total		250

Step 5: Calculate population variance

Population variance is calculated using:

$$\sigma^2 = \Sigma (X - \mu)^2 / N$$

$$\sigma^2 = 250 / 5 = 50$$

Therefore, the population variance is 50 minutes².

Final Answer: Mean = 30 minutes; Population Variance = 50 minutes².

Question 2:

Why Sampling Is Used

A university has 5,000 students. A random sample of 100 students is selected to estimate the average daily study time.

Explain why sampling is used instead of surveying all 5,000 students.

Step 1: Identify the population

Population size (N) = 5,000 students.

The population consists of all students in the university.

Step 2: Identify the sample

Sample size (n) = 100 students.

A random sample is selected so that the sample can provide information about the larger population.

Step 3: Calculate the percentage of students sampled

$$\textbf{Percentage sampled} = \textbf{(Sample size / Population size)} \times \textbf{100}$$

$$= (100 / 5000) \times 100 = 2\%$$

Step 4: Understand why sampling is preferred

• Time saving: Collecting information from 100 students is much faster than collecting information from all 5,000 students.

• Cost saving: Fewer participants generally require fewer resources for data collection and processing.

• Effort reduction: Managing, recording, checking, and analyzing 100 responses is easier than handling 5,000 responses.

• Practicality: A properly selected random sample can provide a useful estimate of the population characteristic.

Conclusion

Sampling is used because surveying all 5,000 students would require considerably more time, cost, and effort. A random sample of 100 students represents only 2%

of the population and can be used to estimate the average daily study time efficiently.

Question 3:

Sample Proportion

From a sample of 200 customers, 150 customers are satisfied with a product.

Calculate the sample proportion of satisfied customers and express it as a percentage.

Step 1: Identify the given values

Total sample size (n) = 200

Number of satisfied customers (x) = 150

Step 2: Use the sample proportion formula

$$\hat{p} = x / n$$

$$\hat{p} = 150 / 200 = 0.75$$

Step 3: Convert the proportion into a percentage

$$\text{Percentage} = \hat{p} \times 100$$

$$= 0.75 \times 100 = 75\%$$

Final Answer

Sample proportion = 0.75

Percentage of satisfied customers = 75%

Question 4:

Covariance and Direction of Relationship

Calculate the covariance between the following variables and determine whether the relationship is positive or negative.

X	Y
2	5
4	7
6	9
8	11

Step 1: Calculate the mean of X

$$\bar{X} = (2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$$

Step 2: Calculate the mean of Y

$$\bar{Y} = (5 + 7 + 9 + 11) / 4 = 32 / 4 = 8$$

Step 3: Calculate deviations and cross-products

X	**Y**	**$X - \bar{X}$**	**$Y - \bar{Y}$**	**$(X - \bar{X})(Y - \bar{Y})$**
2	5	-3	-3	9
4	7	-1	-1	1
6	9	1	1	1
8	11	3	3	9
Total				20

Step 4: Calculate covariance

Using the population covariance formula:

$$\text{Cov}(X, Y) = \Sigma [(X - \bar{X})(Y - \bar{Y})] / N$$

$$\text{Cov}(X, Y) = 20 / 4 = 5$$

Covariance = 5.

Step 5: Interpret the result

The covariance is greater than zero. Therefore, the relationship is positive. When X increases, Y also increases.

Final Answer: Covariance = 5; Relationship = Positive.

Question 5:

Covariance Between Exercise Hours and Calories Burned

The following data represents hours of exercise (X) and calories burned (Y). Calculate the covariance and interpret the result.

Exercise Hours (X)	Calories Burned (Y)
1	200
2	300
3	400
4	500
5	600

Step 1: Calculate the mean of X

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5 = 15 / 5 = 3$$

Step 2: Calculate the mean of Y

$$\bar{Y} = (200 + 300 + 400 + 500 + 600) / 5 = 2000 / 5 = 400$$

Step 3: Calculate deviations and cross-products

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
1	200	-2	-200	400
2	300	-1	-100	100
3	400	0	0	0
4	500	1	100	100
5	600	2	200	400
Total				1000

Step 4: Calculate covariance

$$\text{Cov}(X, Y) = \Sigma\,[(X - \bar{X})(Y - \bar{Y})] / N$$

$$\text{Cov}(X, Y) = 1000 / 5 = 200$$

Therefore, covariance = 200.

Step 5: Interpret the covariance

The covariance is positive because 200 > 0. This means exercise hours and calories burned move in the same direction: as exercise hours increase, calories burned also increase.

Final Answer: Covariance = 200; Relationship = Positive.

Question 6:

Pearson's Correlation Coefficient

Calculate Pearson's correlation coefficient for the following dataset and interpret the strength and direction of the relationship.

Study Hours (X)	Marks (Y)
1	45
2	50
3	55
4	65
5	70

Step 1: Calculate the mean of X

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5 = 15 / 5 = 3$$

Step 2: Calculate the mean of Y

$$\bar{Y} = (45 + 50 + 55 + 65 + 70) / 5 = 285 / 5 = 57$$

Step 3: Calculate deviations and required products

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$	$(X - \bar{X})(Y - \bar{Y})$
1	45	-2	-12	4	144	24
2	50	-1	-7	1	49	7
3	55	0	-2	0	4	0
4	65	1	8	1	64	8
5	70	2	13	4	169	26
Total				10	430	65

Step 4: Write Pearson's correlation formula

$$r = \Sigma\, [(X - \bar{X})\,(Y - \bar{Y})] \,/\, \sqrt{\{\Sigma\,(X - \bar{X})^2 \times \Sigma\,(Y - \bar{Y})^2\}}$$

Step 5: Substitute the calculated values

$$r = 65 / \sqrt{(10 \times 430)}$$

$$r = 65 / \sqrt{4300}$$

$$r = 65 / 65.574$$

$$r \approx 0.991$$

Step 6: Interpret the value

Pearson's correlation coefficient ranges from −1 to +1.

Value of r	General interpretation
Close to +1	Strong positive relationship
Close to 0	Weak or no linear relationship
Close to −1	Strong negative relationship

Here, r ≈ 0.991, which is very close to +1. Therefore, the dataset shows a very strong positive linear relationship between study hours and marks.

In practical terms, students with more study hours tend to have higher marks in this dataset.

Final Answer: Pearson's correlation coefficient r ≈ 0.991; Relationship = Very strong positive correlation.

SUMAARY

Question	Topic	Final Answer
1	Mean	30 minutes
1	Population Variance	50 minutes^2
2	Sampling	Used to save time, cost, and effort
3	Sample Proportion	0.75 = 75%
4	Covariance	5
4	Relationship	Positive
5	Covariance	200
5	Relationship	Positive
6	Pearson Correlation	r ≈ 0.991
6	Relationship	Very strong positive